

power.

28

C

From photoelectric equation, $eV_s = hf - \Phi$

Using the same ultraviolet source is used, the energy of each photon remains the same, thus the stopping potential remains the same.

Lower intensity implies that the rate of photons incident on the metal is less. The rate of electron emission is less, and the rate of electrons reaching the collector plates also decreases. Thus, the new saturation current is less than the original.

29

A

When the potential difference is decreased, the energy of the most energetic x-ray photon